



Jekaterina Truskina

PhD in Biology | Data Scientist and Computer Vision engineer

Data Scientist specialized in biological imaging and computer vision, with dual expertise in biology and deep learning. I develop advanced image analysis methods and robust AI models to study complex biological systems, covering the full pipeline—from data labeling and augmentation to training, fine-tuning, deployment, and client interface development.

CONTACT

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Goncelin (38), France Permis B

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SKILLS

Data Science, IA and Computer Vision

- **Computer Vision:** YOLO, Vision Transformers, Cascade RCNN, Autoencoders, feature extraction, segmentation
- **AI Pipeline:** Image labeling, data augmentation, model optimization, server/edge deployment, automation, integration with data pipeline
- **Image Processing:** Advanced pipelines: segmentation, filtering, edge detection
- **Languages & Frameworks:** Python (pandas, numpy, PyTorch, TensorFlow, OpenCV, scikit-learn), R, Linux
- **Machine Learning:** Regression, KNN, SVM, XGBoost, Clustering (DBSCAN/HDBSCAN)
- **Data Bases:** SQL, noSQL
- **Application Development:** build interactive client interfaces via FastAPI and Streamlit

Scientifique Skills

- **Microscopy:** confocal, scanning and transmission electron, Raman
- **Plant Genetics and Epigenetics**
- **Plant Transcriptomics**
- **Plant Pathology**
- **Plant Biochemistry**
- **Molecular Biology**
- **Histology**

Languages

- English: C1 French: C1
- German: C1

WORK EXPERIENCE

Data Scientist and R&D Engineer

Oberon Sciences, Villard-Bonnot, France

- **Design and training of object detection and segmentation models** (YOLO, RF-DETR, Cascade R-CNN, Autoencoders, Feature Extraction) from images captured by optical detectors
- **Comparative benchmarking of architectures** based on accuracy (mAP), latency (FPS), and memory usage
- **Model optimization for edge/server deployment** (model size reduction, FPS improvement, CPU/GPU inference optimization)
- **Development of a complete AI pipeline:** image labeling, data augmentation, fine-tuning, automated deployment, database integration, and API development for industrial applications

2026
-
2024

Postdoctoral Scientific Researcher

CNRS, ENS de Lyon, France and IBP Halle, Germany

Scientific research in plant reproduction and biochemistry

2023
-
2018

- Study of reactive oxygen species in pollen and seeds
- Analysis of the composition and biosynthesis of the pollen wall
- Research on pollen development: roles of membrane receptors and peptides

PhD Candidate in Biology

ENS Lyon, France and University of Nottingham, UK

Scientific research in plant genetics:

2018
-
2014

transcriptomic and epigenetic regulation in plant hormone signaling pathways

Research Assistant

University of Göttingen Germany

2013

Scientific research on plant pathogens: fungal infections

EDUCATION

Data Scientist and Data Analyst

M2i Formation, Lyon, France

2024

PhD in biologie

ENS Lyon, France and University of Nottingham, UK

2018

Master in microbiology and biochemistry

University of Göttingen Germany

2012

Publications

1. Doll NM, **Truskina J**, Ingram G. Functional and developmental convergence in the reproductive "nurse cells" of flowering plants. *C R Biol.* 2023 May 30;346:45-54. doi: 10.5802/crbiol.119.
2. **Truskina J**, Boeuf S, Renard J, Andersen TG, Geldner N, Ingram G. Anther development in *Arabidopsis thaliana* involves symplastic isolation and apoplastic gating of the tapetum-middle layer interface. *Development* 2022 149, dev200596.
3. Doll NM, Berenguer E, **Truskina J**, Ingram G. AtEXT3 is not essential for early embryogenesis or plant viability in *Arabidopsis*. *New Phytol* 2022 236, 1629–1633. 10.1111/nph.18452.
4. **Truskina J**, Brück S, Stintzi A, Boeuf S, Doll NM, Fujita S, Geldner N, Schaller A, Ingram GC. A peptide-mediated, multilateral molecular dialogue for the coordination of pollen wall formation. *Proc Natl Acad Sci U S A.* 2022 119, e2201446119. <https://doi.org/10.1073/pnas.2201446119>.
5. **Truskina J**, Han J, Chrysanthou E, Galvan-Ampudia CS, Lainé S, Brunoud G, Macé J, Bellows S, Legrand J, Bågman A-M, Smit ME, Smetana O, Stigliani A, Porco S, Bennett MJ, Mähönen AP, Parcy F, Farcot E, Roudier F, et al. A network of transcriptional repressors modulates auxin responses. *Nature.* 2021 Jan;589(7840):116–9.
6. **Truskina J**, Vernoux T. The growth of a stable stationary structure: coordinating cell behavior and patterning at the shoot apical meristem. *Curr Opin Plant Biol.* 2018 Feb;41:83–8.
7. Orosa-Puente B, Leftley N, von Wangenheim D, Banda J, Srivastava AK, Hill K, **Truskina J**, Bhosale R, Morris E, Srivastava M, Kümpers B, Goh T, Fukaki H, Vermeer JEM, Vernoux T, Dinneny JR, French AP, Bishopp A, Sadanandom A, et al. Root branching toward water involves posttranslational modification of transcription factor ARF7. *Science.* 2018 Dec 21;362(6421):1407–10.
8. Bhosale R, Giri J, Pandey BK, Giehl RFH, Hartmann A, Traini R, **Truskina J**, Leftley N, Hanlon M, Swarup K, Rashed A, Voß U, Alonso J, Stepanova A, Yun J, Ljung K, Brown KM, Lynch JP, Dolan L, et al. A mechanistic framework for auxin dependent *Arabidopsis* root hair elongation to low external phosphate. *Nat Commun.* 2018 Apr 12;9(1):1409.
9. Reusche M, **Truskina J**, Thole K, Nagel L, Rindfleisch S, Tran VT, Braus-Stromeier SA, Braus GH, Teichmann T, Lipka V. Infections with the vascular pathogens *Verticillium longisporum* and *Verticillium dahliae* induce distinct disease symptoms and differentially affect drought stress tolerance of *Arabidopsis thaliana*. *Environmental and Experimental Botany.* 2014 Dec 1;108:23–37.
10. Reusche M, Klásková J, Thole K, **Truskina J**, Novák O, Janz D, Strnad M, Spíchal L, Lipka V, Teichmann T. Stabilization of cytokinin levels enhances *Arabidopsis* resistance against *Verticillium longisporum*. *Mol Plant Microbe Interact.* 2013 Aug;26(8):850–60.
11. Reusche M, Thole K, Janz D, **Truskina J**, Rindfleisch S, Drübert C, Polle A, Lipka V, Teichmann T. *Verticillium* infection triggers VASCULAR-RELATED NAC DOMAIN7-dependent de novo xylem formation and enhances drought tolerance in *Arabidopsis*. *Plant Cell.* 2012 Sep;24(9):3823–37.